Roll No.....

Mid Semester Examination March 2025

Name of the Program: ***B.Tech*** *Semester:* **IV**

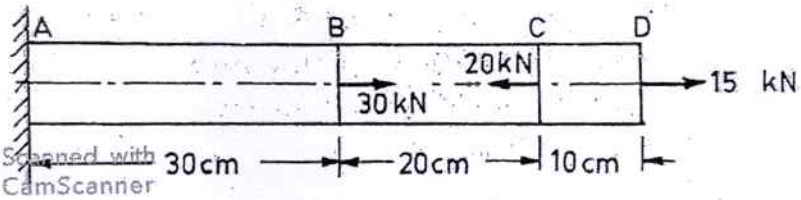
Name of the Course: **Mechanics of Materials** *Course Code:* **TME406**

Time: 90 Minutes Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	**CO1**
(a)	Explain the following terms: (i) Elastic Constants (ii) Tensile test diagram for steel	
	OR	
(b)	A steel bar of 10 mm diameter is subjected to an axial load of 12 kN. If the change in diameter is found to be 0.0022 mm, determine the Poisson's ratio and the modulus of elasticity and the bulk modulus. Take G=78 GPa.	
Q2	(10 marks)	**CO1**
(a)	Deduce expression to determine the elongation of a bar of tapering circular section.	
	OR	
(b)	A steel bar of 30 mm diameter is loaded as shown in Figure. Determine the stress in each portion and the total elongation. E = 210 GPa.  	
Q3	(10 marks)	**CO1, CO2**
(a)	Rails are laid such that there is no stress in them at 24° C. If the rails are 32 m long. Determine: (i) Stress in the rail at 80° C, when there is no allowance for expansion (ii) Stress in the rail at 80° C, when there is expansion allowance of 8mm per rail. (iii) The expansion allowance for no stress in the rail at 80°C. Take: Coefficient of linear expansion = 11 x 10^(-6) /°C, E= 205 GPa	
	OR	
(b)	(i) Deduce the expression for elongation of a bar of rectangular section due to self-weight. (ii) A steel wire of 8-mm diameter is used to lift a weight of 1.5 kN at its lowest end. The density of the wire material is 8000 kg/m³. Determine the elongation of the wire if the length of the wire is 100m. E=205 GPa	

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Q4	(10 marks)	**CO2**
(a)	What is Mohr's stress circle? How the Mohr's circle is constructed?	
	OR	
(b)	A rectangular block is subjected to two perpendicular stresses of 10MPa tension and 10MPa compression. Determine the stresses on inclined plane at (i) 30° (ii) 45° (iii) 60° with the plane of compressive stress.	
Q5	(10 marks)	**CO2**
(a)	Derive the expression for stresses on an inclined plane in a body subjected to a bi-axial stress condition.	
	OR	
(b)	Draw Mohr's stress circle indicating the radius and centre point coordinate for given stress conditions. (i) Pure shear (60 MPa) (ii) Hydrostatic stresses (120 MPa) (iii) Normal stress in X- direction 120 MPa(tensile) and Normal stress in Y-direction 60 MPa(tensile) (iv) Normal stress in X- direction 120 MPa(tensile) and Normal stress in Y-direction 60 MPa(compressive) (v) Normal stress in X- direction 120 MPa(tensile) and Normal stress in Y-direction 60 MPa(tensile) with 60 MPa shear stress	